Augmenting Expert Elicitation With LLMs for Building Causal Models to Aid Decision Making

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Abstract*—*Objective: **This study examines whether Large Language Models (LLMs) can serve as a supplement to traditional expert input when extracting structured causal knowledge, for the case study of decision support in medical diagnostics. Materials and *Methods:* We compare LLM-generated causal structures against several benchmark causal graphs created for medical diagnostic datasets. To reduce prompt-based variance and enhance reliability, we introduce a novel multi-prompt strategy where each pairwise variable query is tested using seven distinct causal linking verbs. Final edge inclusion is determined by majority voting across these variants. The resulting graphs are evaluated using standard structural similarity measures, such as Structural Hamming Distance, and Total Edge Recall Error. LLMs are able to reconstruct a substantial portion of the true causal structure, improving performance for aggregated responses from multiple prompt variations. Averaging across several prompts mitigated biases introduced by individual phrasing and reduced dependence on any single causal template, confirming that ensemble prompt strategies improve robustness. *Conclusion:* LLMs show promise as aids to human experts, particularly in settings where expert input is sparse or delayed. Our majority vote multi prompt technique provides a scalable method to reduce linguistic sensitivity in causal elicitation, validated on a broader LLM augmented framework.**

***Index Terms*—Large language models, bayesian networks, causal modelling, expert elicitation, causal graph, benchmark, directed acyclic graph, structural similarity.**

I. Introduction

IN MANY domains, particularly healthcare and public health, researchers seek to answer questions about cause and effect, such as: What is the effect of a treatment? What factors influence disease progression? However, such questions cannot be answered from observational data alone or even from the statistical distributions that describe the data. An approach called causal inference [1], [2] shall be applied, using the representations such as a probabilistic causal graphs [3], including Structural Equation Model (SEM) and Bayesian Network (BN).

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Building these graphs is difficult [3], especially in complex medical fields. Traditionally, we rely on expert elicitation, in which specialists provide their knowledge to build the model [4], [5], [6]. While effective, this process is slow, expensive, and hard to scale. It requires multiple interviews and training experts in probability, making it difficult to use when rapid model development is needed.

Recently, Large Language Models (LLMs) have become a new tool for causal discovery [7], [8], [9], [10], [11]. LLMs, such as GPT-4o-mini, are trained on massive and diverse datasets, including scientific literature, textbooks, and web content. This training allows them to capture a wide range of domain knowledge and infer plausible relationships between concepts based on variable names, definitions, or context. Researchers have begun using LLMs as "virtual experts" to propose causal links, answer conditional independence queries, and even generate full or partial DAGs to support BNs construction.

A critical limitation of using LLMs for causal discovery is the high sensitivity of their outputs to prompt formulation [12], [13]. Small changes in how a question is phrased can lead to different results. For example, asking whether "X causes Y" may yield a different LLM response than asking whether "X leads to Y," "X influences Y," or "X is associated with Y." Because LLMs are trained on diverse web and scientific text where these terms are used inconsistently, the resulting models are often unstable and unreliable if only one phrasing is used.

Our contribution addresses this instability directly. Instead of relying on a single prompt, we propose a novel framework that uses a diversity of seven causal linking verbs for every variable pair. The novelty of our work lies in the following:

- We systematically statistically explore how different causal verbs affect the model's logic.
- We introduce a consensus-based method to combine these results. We effectively filter out linguistic noise and hallucinations caused by a single verb using a majority-voting approach.
- We demonstrate that this ensemble approach increases the stability and accuracy of the generated structures across five diverse medical benchmarks, providing a more stable method for LLM-driven causal discovery.

This paper is structured as follows. Section II reviews related work in causal discovery, expert elicitation, and LLM-based knowledge extraction. Section III formulates the problem of using LLMs to discover causal structure. Section IV presents the methodology, including dataset selection, prompt design, and evaluation metrics. Section V describes the experimental

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results and analysis. Section V investigates verb sensitivity and lexical framing effects in LLM-based causal inference. Section VII discusses the implications, limitations, and opportunities for future research. Section VIII concludes the paper.

II. Related Work

Causal discovery using probabilistic graph models seeks to identify relationships among variables, often represented as a Directed Acyclic Graph (DAG). More advanced models, such as Bayesian Networks (BNs), allow to perform causal inference on graphs [1], [2]

The DAG becomes a BN once each node (variable) in the graph is assigned a conditional probability table that reflects the distribution of the variable.

Beyond causal modelling, recent work has proposed advanced graph-based frameworks for medical and data-intensive tasks. For example, the Tight-fitting Graph Inference Network (TF-GIN) [14] graph structure for table-based fact verification, and the Medical Knowledge Graph (MKG) [15] learning framework that applies Graph Neural Networks to support clinical decisions, such as predicting medication combinations. Although effective, these approaches follow paradigms that differ from our focus. They typically rely on multi-relational or undirected graphs to capture associations or verify data. In contrast, our work centers on eliciting DAGs for BNs. A DAG imposes strict topological constraints to enable valid probabilistic inference and causal reasoning.

The first and often most critical step in causal analysis is constructing a DAG that reflects the underlying causal structure. These graphs encode assumptions about causal pathways, confounding variables and mediators. Confounding variables are factors that influence both the cause and the effect, potentially biasing the observed relationship. Mediators are variables that lie on the causal pathway, transmitting the effect of one variable onto another.

Approaches to constructing DAGs include:

- *Data-driven methods*, which infer causal structure directly from data; these include: (a) Score-based approaches (e.g., Bayesian Information Criterion scoring [16]); (b) Constraint-based approaches (e.g., the Peter–Clark [PC] algorithm [17]); and (c) Hybrid approaches, which combine elements of both (e.g., Max-Min Hill-Climbing [18])
- *Expert elicitation*, which relies on domain knowledge to define or refine causal structures [4], [5], [6].
- *LLM-based methods*, a new approach that uses LLMs to suggest causal edges through prompting [7], [11].

Traditional data-driven methods for learning causal structures face significant challenges, as the number of possible structures grows exponentially with the number of variables, making a comprehensive search intractable [3]. Additionally, knowledge about causal relationships is often incomplete, and the rapid growth of research and data makes it increasingly difficult to manually synthesize the evidence required to build accurate and comprehensive DAGs [19].

To address this problem, researchers are increasingly relying on expert knowledge to inform the search for plausible structures. This process, called expert elicitation, involves gathering judgments from domain experts to inform or constrain the causal model.

In [6], elicitation approaches range from unstructured to structured. Structured approaches provide better conditions for expert judgments (e.g., reference protocols, challenge clustering mechanisms, statistical significance techniques, and machine learning/reasoning), resulting in better control of cognitive biases and process transparency.

Structured elicitation protocols, developed over decades [4], [5], [20], aim to mitigate cognitive biases, such as confirmation bias, framing effects, and ambiguity, thereby enhancing the reliability of expert input.

Expert knowledge can be incorporated as hard constraints, such as requiring or forbidding specific edges, enforcing variable orderings, or applying type-based restrictions [21], [22], [23]. These strategies can effectively reduce the search space but may exclude the true structure if the provided knowledge is inaccurate.

In the process of constructing BNs [24], [25], this may include identifying the presence or absence of edges, specifying variable orderings, or providing prior distributions over causal effects.

However, despite best practices, expert-generated DAGs and BNs are not always accurate and may miss critical pathways, including hidden confounders or indirect effects [26].

Recent studies explore LLMs as a novel approach to integrating soft expert knowledge into causal discovery [7], [8], [9], [10], [11]. Because LLMs are trained on a large and diverse text corpus, they encode a wide range of factual and commonsense information. This makes them particularly valuable in settings where domain-specific data is limited. LLMs can support reasoning over variable names, descriptions, and metadata to suggest plausible causal relationships [27], [28], [29].

LLMs have been used both to generate causal predictions directly [13], [28] and to produce priors for score-based algorithms [30]. [31] proposes a divide-and-conquer framework that uses weak prior knowledge inferred from LLMs for stable, data-independent variable partitioning. A score-based refinement step is then used to eliminate spurious causal links. [32] introduces a harmonized prior that combines precision- and recall-oriented reasoning modes derived from LLMs. This prior is integrated into structure learning algorithms (score-based, constraint-based, and gradient-based) via plug-and-play strategies. While earlier work emphasized improving overall graph accuracy, newer research focuses on task-specific metrics, such as identifying correct causal orderings that are critical for valid effect estimation [9].

Regardless of their advantages, LLMs are highly sensitive to the phrasing of prompts. Slight differences in wording, such as “Does A cause B?” versus “Is B affected by A?”, can produce different results, particularly when identifying causal directions [9]. Prompting strategies, such as chain-of-thought reasoning, help improve logical consistency by encouraging step-by-step deductions [33].

In causal tasks, this sensitivity is significant. In [13], it was found that LLMs answered conditional independence queries

more accurately when given structured prompts that included domain-specific language, specifications, and explanations.

Moreover, LLMs' training data include a mix of formal scientific texts and informal language, which contributes to inconsistent interpretations of causal phrases. Terms like `associated with`, `correlates with`, and `leads to` may be used interchangeably, despite their different meanings in causal analysis [9].

Prompt granularity also affects LLM performance. [34] introduces a triplet-based approach, where the model reasons about causal direction between three variables at a time. This method outperformed pairwise prompts at producing acyclic orderings and at reducing topological divergence, particularly in low-data settings. However, this approach requires more queries as the number of variables increases, highlighting a trade-off between accuracy and scalability.

To address these limitations, hybrid approaches have been proposed. For example, [34] combines LLM-derived causal orderings with existing algorithms, such as the constraint-based Peter Clark and the score-based Context-Aware Multi-Modal Learner, achieving better performance than either method alone.

Our approach shares several fundamental similarities with existing literature. We utilize classical prompt formulation strategies, which have been identified in literature reviews as providing the best baseline performance for causal tasks. Furthermore, our goal aligns with the broader research trend of using LLMs to DAGs for structural modelling.

However, the key divergence lies in the role of the LLM and how its outputs are consolidated. Hybrid methods [30], [31], [32], [34] treat the LLM as a prior-generator refined by a downstream score- or constraint-based algorithm, and therefore still require observational data. ALCM-style approaches [8] interleave LLM querying with algorithmic search, again data-dependent. Our method is purely elicitation-based without data-driven refinement. Instead, we aggregate responses across multiple semantically equivalent but lexically varied prompts via majority voting. This directly neutralises the per-verb linguistic bias that prior work acknowledges but does not quantitatively address, yielding a more stable, data-free consensus structure than single-prompt or hybrid strategies.

III. Problem Formulation

Given a set of variables $X_1, X_2, \ldots, X_n$, our goal is to construct a causal graph that captures the direct causal relationships between these variables. We represent this causal structure as a DAG, where nodes represent variables, and directed edges represent causal relationships between pairs of variables. Formally, a causal graph is a DAG $G = (V, E)$, where: $V = \{X_1, X_2, \ldots, X_n\}$ is the set of variables, and $E \subseteq V \times V$ is the set of directed edges.

The problem of constructing such a causal graph involves identifying which edges (causal relationships) should be present in the graph and determining their direction. Typically, expert knowledge is required to assess whether a variable X_i causes another variable X_j, which we denote by $X_i \to X_j$.

In this paper, we propose to use LLMs to elicit causal relationships. We define the expert elicitation task as follows:

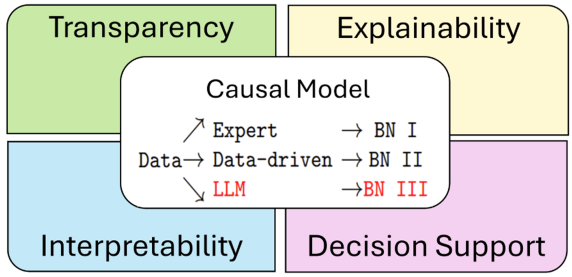


Fig. 1. Diagram to illustrate three common approaches to constructing BNs: Expert elicitation (BN I), Data-driven methods (BN II), and LLM-based methods as an additional method that supports expert elicitation (BN III). While traditional approaches contribute domain knowledge or data quality, LLMs enhance the explainability, interpretability, transparency, and decision support of the resulting causal models by making the structure and reasoning behind the graph more accessible and manageable.

- For each pair of variables X_i and X_j, we pose a query $Q_{i,j}$ asking the LLM whether X_i causes X_j, and if so, the direction of causality (i.e., $X_i \to X_j$ or $X_j \to X_i$).
- The query is formalized as a variable with two possible outcomes (True or False) for each state:
  - $Q_{i,j}$: a direct causal link from X_i to X_j exists (i.e., $e_{i,j} = 1$),
  - $Q_{j,i}$: a direct causal link from X_j to X_i exists (i.e., $e_{j,i} = 1$).
- The model returns the prior (log-probability) for each output token to verify the endpoints.

This process produces a set of directed edges that represent the LLM's causal graph G_{LLM}. To ensure the validity of the generated graph, we compare it with ground truth graphs (where available) and assess its coherence using various graph-based metrics. A critical challenge in this setup is the sensitivity of LLM outputs to the choice of causal linking verbs in the query for the prompt.

To mitigate this problem, we propose a multi-verb strategy as follows:

- for each variable pair (X_i, X_j), issue multiple prompts using seven different causal linking verbs, and
- determine edge inclusion through majority voting.

The main objective of this study is to investigate whether LLMs can be used to extract causal relationships and construct valid causal graphs reliably. Specifically, we aim to:

- Aggregate results across verb variants to mitigate lexical bias and surface robust relationships.
- Provide verb-sensitive analysis to explore how the choice of causal linking verb formulation impacts the output.
- Evaluate the quality of LLM-generated causal graphs using various graph metrics, such as Structural Hamming Distance (SHD), True Edge to Negative Edge (TENE), and True Edge to Reverse Edge (TERE).
- Compare the LLM-generated graphs against expert-constructed ground truth graphs.

We propose an LLM-augmented approach to constructing BN (BN III) that differs from the expert-derived (BN I) and data-driven (BN II) methods (see Fig. 1). This approach offers advantages in terms of explainability and interpretability. While expert-derived BNs (BN I) provide high interpretability

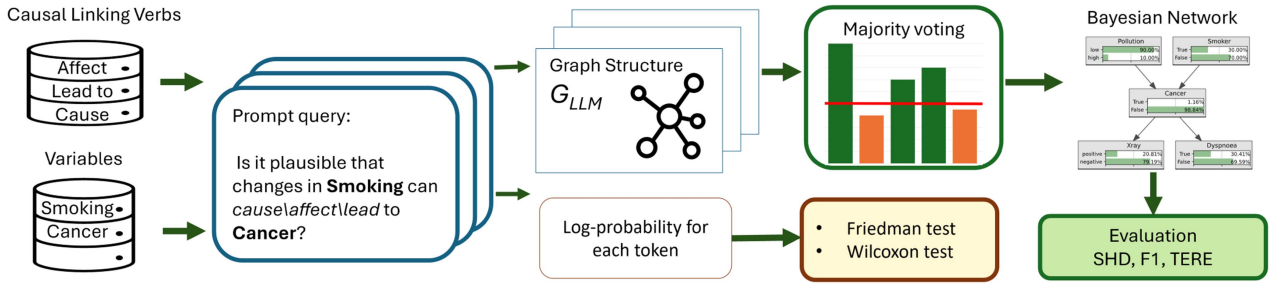


Fig. 2. LLM-based Methodology for Causal Graph Construction and Evaluation. This workflow illustrates how LLMs are prompted with causal queries combining domain concepts (e.g., Smoking, Cancer) and varied causal linking verbs (e.g., cause, lead to, affect). For each pair, LLM responses are used to construct a graph structure G_{LLM}, with edge decisions aggregated via majority voting. Token-level log-probabilities are also extracted to enable statistical comparisons across prompt formulations using the Friedman and Wilcoxon tests. The final graph is evaluated against the ground truth using structural and performance metrics.

grounded in domain expertise (as shown in the related work [35], and data-driven BNs (BN II) offer precision (as explained in [36]), the proposed LLM-based method (BN III) leverages language-based reasoning to generate causal structures that are transparent to enhance decision support capabilities.

In particular, the key explainability features include:

- *Transparent Edge Generation:* Each causal relationship proposed by the LLM can be traced back to patterns in its training corpus, with the ability to generate natural language explanations for edge propositions (e.g., linking "smoking" to "lung cancer" based on patterns in medical literature).
- *Multi-prompt Consensus:* The method reduces linguistic bias while surfacing robust, consensus relationships, by aggregating results across varied causal linking verbs formulations (e.g., "causes", "increases risk of", "leads to").

The described approach addresses critical gaps in current BN construction methods by balancing the scalability of data-driven approaches with the transparency of expert-driven methods. In the current study, we validate only the automated elicitation step driven by the LLM. While LLMs can automate and scale the elicitation process, human experts remain essential to validate and refine the resulting structure. Designed to augment rather than replace domain experts, this framework aims to facilitate development of a future human-in-the-loop validation framework.

Through an experimental study, we attempt to answer the following research questions:

- How sensitive are LLMs to linguistic variability in prompt formulation and to what extent can such differences systematically distort the causal structures they generate?
- Can aggregating outputs across causal linking verbs choices mitigate lexical bias?

In future work, domain experts would review and adjust the LLM-generated DAG to ensure domain validity, interpretability, and alignment with real-world knowledge.

IV. Approach and Methodology

This section provides the methodology for using LLMs for expert elicitation in constructing causal models. Our objective is to determine whether LLMs can serve as a reliable source of expert knowledge, independently or in combination with human experts, particularly in healthcare contexts where domain expertise is limited or costly. The general scheme of the project is shown in Fig. 2.

The research workflow begins by framing causal queries using pairs of domain concepts and multiple causal linking verbs to test their effect on LLM responses. These outputs are used to construct a graph, with edge decisions aggregated via majority voting. Token log-probabilities are extracted and compared across prompt formulations to assess how verb choice influences LLM outputs and to identify potential lexical biases in its outputs. Finally, the constructed graph is evaluated against the ground truth using structural and performance metrics.

We implemented this procedure using the LangChain framework for prompt formatting and the OpenAI API for model inference with GPT-4o-mini at a fixed temperature of 0.

A temperature of 0 minimizes randomness, forcing the model to always choose the most likely next token. This is essential for our experiments because it ensures that the differences we see in the causal graphs are caused by our specific linguistic prompts, not by random variations in the LLM's output. Log-probabilities were parsed from model response metadata, and customized functions were used to extract relevant tokens and their log-probabilities.

A. Domain and Dataset Selection

We selected a suite of benchmark causal datasets [37] widely recognized in the healthcare and medical probabilistic graphical modelling literature, specifically the Asia, Cancer, and Medical Diagnostic networks. As these datasets provide well-established ground-truth causal structures, they serve as a gold standard for evaluating the fidelity of our elicited models. To ensure contemporary relevance, we augmented this selection with modern datasets concerning Covid-19 [38] and Endometrial cancer (Endorisk) [39] (Table I)

To maximize clarity and prevent misinterpretation of the nodes in datasets, we include for each variable:

- The name of the node (e.g., Cancer, Smoker, X-ray Result), which describes the concept or medical entity under evaluation.
- The possible states of that node (e.g., Yes/No, Present/Absent, or domain-specific labels such as

TABLE I
SUMMARY OF BENCHMARK DATASETS USED FOR EVALUATING LLM-ELICITED CAUSAL GRAPHS

Dataset	# of Nodes	# of Edges	Description
Asia	8	8	A simple medical diagnosis network modeling respiratory disorders in patients who have recently travelled to Asia.
Cancer	5	4	A small network illustrating potential causal factors and effects related to the presence of lung cancer.
Medical Diagnostic	10	9	A small dataset for the detection of cardiovascular diseases that includes symptoms and diseases
Covid-19	11	20	A contemporary causal model representing risk factors, clinical presentations, and outcomes of infection.
Endorisk	18	37	A complex clinical model used for predicting metastasis in endometrial cancer based on preoperative data.

TABLE II
PROMPT TEMPLATE FOR CAUSAL DISCOVERY

LLM Prompt Template for Discovery Causal Relationship
`System`: You are a reasoning assistant tasked with identifying whether a *causal* relationship exists between two medical-related variables. Each variable has discrete states, listed in parentheses. Assume that variables represent conditions, risk factors, symptoms, or diagnostic test results related to a single patient. Causality should reflect real-world biomedical reasoning: for example, a disease might cause a symptom, or a risk factor might cause a disease. Only return 'True' if changing the first variable is likely to lead to a change in the second. *Return only 'True' or 'False'*. Do not confuse correlation with causation." `Human`: Consider the following two variables: - {cause}: ({cause_states}) - {effect}: ({effect_states}) Is it plausible that changes in '{cause}' can {causal_verb} '{effect}'? Answer only with 'True' or 'False'.

Mild/Moderate/Severe, or numeric categories where applicable).

We focused on healthcare as a stress test for LLM-based discovery, given the field's demand for high-fidelity terminology and its frequent contradictions in expert literature. This choice supports our hypothesis that the framework is fundamentally transferable. Because the core linguistic sensitivity issue is present whenever specialized language describes interactions, our strategy should remain effective across diverse disciplines.

B. Causal Elicitation

To extract causal knowledge from the LLM, we employed a carefully engineered [40], [41] prompt template designed to guide the model toward biomedical-style causal reasoning (Table II). We used a few-shot ChatPromptTemplate instruction format where the system role set the context and expectations, and the user message posed a directed causal query between a pair of medical-related variables. Each variable represents a condition, risk factor, symptom, or diagnostic test outcome associated with a single patient. The system message explicitly stated the criteria for causality: a change in the first variable should be likely to cause a change in the second. This ensures alignment with real-world healthcare logic, such as "smoking increases the risk of lung cancer" or "pneumonia causes shortness of breath." The model was instructed to return only a binary decision: `True` or `False`.

To assess the robustness and linguistic sensitivity of the LLM's causal reasoning, we introduce a novel multi-verb elicitation strategy. For every ordered pair of variables (X_i, X_j), we generated multiple prompts, each using one of the causal linking verbs. This design enables us to probe whether the model's causal judgment is stable across different linguistic framings, which has not been systematically explored in prior works. For each variable pair (X_i, X_j), we aggregated the binary responses returned by the model across all verb formulations. The final causal connections for graph G_{LLM} were assigned by majority vote, representing the most consistent causal judgment across variations. This aggregation reduces the impact of prompt phrasing bias.

Let us use the set of variables V in a given dataset. We iterated over all unique pairs (X_i, X_j), where $i \neq j$, and for each pair and verb, we issued a formatted prompt to the LLM. For every prompt, we recorded:

- The pair of variables (cause, and effect),
- The causal linking verbs used,
- The model's binary response (`True` or `False`),
- The log-probability of the returned token.

Log-probabilities were converted to probabilities through an exponential transformation. Only edges with affirmative responses (`True`) were added in set E as directed edges. If there were bidirectional connections, we would spare the connection with the highest probability, where the LLM was more confident in generating tokens.

To represent the LLM-derived causal graph in a compact form, we constructed a binary adjacency matrix $\hat{A} \in \{0,1\}^{n \times n}$. Let $D \subseteq E$ denote the set of directed edges where the model affirmed a causal relationship (i.e., responses labelled as `True` after aggregation across verbs). Each adjacency matrix entry $\hat{A}_{ij} = 1$ if $(X_i, X_j) \in D$, and $\hat{A}_{ij} = 0$ otherwise. This matrix was constructed by mapping the directed causal edges implied by the LLM's answers across all variable pairs, preserving the order of input nodes. The resulting matrix serves as a direct representation of the LLM's inferred causal structure.

To improve clarity and ensure a minimal DAG suitable for BN inference, we applied transitive reduction. In a BN, the condition requires each node to be independent of its non-descendants given only its direct parents. A redundant edge violates this parsimony. It implies a direct causal influence and unnecessarily expands the conditional probability table of the child node. This step ensures that each node's parent set encodes only direct causes, keeping conditional probability tables compact and consistent with BN parsimony assumptions. Transitive reduction removes such edges, yielding an adjacency matrix $\hat{A}'$ that preserves all essential causal pathways while satisfying the conditional independence structure required for valid probabilistic inference.

C. Evaluation Metrics

To assess the quality of the G_{LLM}, we compare the predicted adjacency matrix $\hat{A}$ with the ground truth matrix A using several graph similarity metrics [9], [30]. Our evaluation focuses on

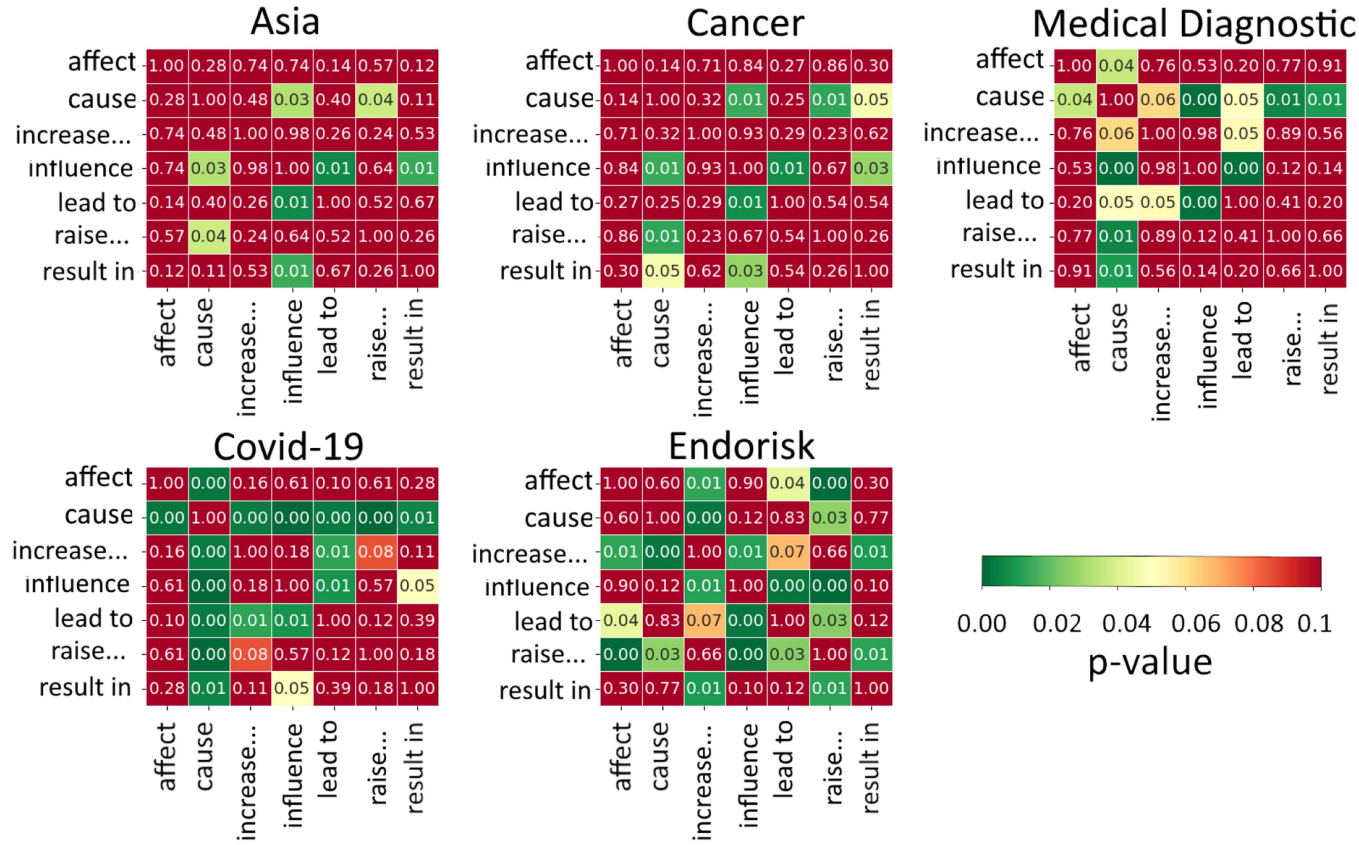


Fig. 3. Pairwise p-values between causal linking verbs across Asia, Cancer, Medical Diagnostic, Covid-19, and Endorisk datasets. The p-value below 0.05 is highlighted in bold, which represents a statistically significant difference. The verbs `increase` and `raise` were used as shortened versions of the full prompts `increase the chance of` and `raise the risk of`.

both structural correctness and directional fidelity. We normalize each metric to enable fair comparison across datasets of different sizes. In all cases, lower values indicate better performance, meaning closer alignment with the true causal structure.

a) Structural Hamming Distance (SHD): measures the total number of mismatched entries between the predicted and ground truth adjacency matrices:

$$\text{SHD}(A, \hat{A}) = \sum_{i,j} [A_{ij} \neq \hat{A}_{ij}]$$

This includes all false positives, false negatives, and reversed edges. SHD is a total error count. It shows the total number of incorrect edge decisions. To normalize SHD, we divide by the total number of matrix elements (N):

$$\text{Normalized SHD} = \frac{\text{SHD}(A, \hat{A})}{N}$$

b) True Edge to Negative Edge (TENE): quantifies the proportion of true causal edges from the ground truth that are missing in the predicted graph. TENE measures missing information. It looks only at the real causal links that actually exist and shows what percentage of them the LLM failed to find:

$$\text{TENE}(A, \hat{A}) = \sum_{i,j} [A_{ij} = 1 \wedge \hat{A}_{ij} = 0]$$

$$\text{Normalized TENE} = \frac{\text{TENE}(A, \hat{A})}{\sum_{i,j} [A_{ij} = 1]}$$

c) True Edge to Reverse Edge (TERE): captures the proportion of true edges that are incorrectly reversed in the predicted graph. TERE measures direction errors. It counts how many times the LLM correctly identified that two things are related but got the "cause and effect" backward:

$$\text{TERE}(A, \hat{A}) = \sum_{i,j} [A_{ij} = 1 \wedge \hat{A}_{ji} = 1]$$

$$\text{Normalized TERE} = \frac{\text{TERE}(A, \hat{A})}{\sum_{i,j} [A_{ij} = 1]}$$

TENE penalizes edge omissions, while TERE captures directional errors, both of which are critical for evaluating the correctness of a causal graph.

V. Verb Sensitivity Analysis in Causal Inference

Causal reasoning in healthcare critically depends on how relationships between variables are expressed, as mentioned in the previous section. The verbs used to probe causal relationships significantly influence interpretation in natural language settings [42], [43] In this section, we examine how linguistic variability in prompt formulation affects LLM-inferred causal structures. We focus on the following research question: *How*

sensitive are LLMs to linguistic variability in prompt formulation, such as synonym choice, syntactic framing, or verb semantic, and to what extent can such differences systematically distort the causal structures they generate?

We split this research question into two complementary subquestions:

1) Do subtle changes in phrasing in the LLM prompts lead to significantly different causal models, and to what extent does linguistic variability influence the causal relationships in the models inferred by LLMs?
2) Do such biases lead to systematic distortion of the derived causal structures?

We conducted experiments using the five benchmark datasets to evaluate the reliability of causal relationships extracted via different verbal prompts. For each DAG, we iterated through all possible ordered pairs of variables (X_i, X_j), where (X_i) is a candidate cause and (X_j) is a potential effect.

Causal linking verbs communicate the strength, direction, and type of relationship between variables [44], [45] In healthcare, where causality is often probabilistic and multifactorial, the difference between strong, deterministic verbs (e.g., `cause`, `lead to`) and probabilistic or contributory verbs (e.g., `influence`, `increase the chance of`) is essential. Choosing the appropriate verb mitigates the risk of introducing interpretability errors or bias. The right selection enhances alignment with medical knowledge, improves interpretability, and supports more robust causal structure learning [11].

Following a systematic review of the literature, we selected a set of seven causal expressions: `cause`, `affect`, `influence`, `lead to`, `result in`, `raise the risk of`, and `increase the chance of`. These were chosen based on the following criteria:

- *Diversity of strength and implication:* Verbs may suggest deterministic, strong directional causality, or imply softer, more probabilistic relationships.
- *Common usage in scientific and health literature:* All selected verbs are frequently found in academic writing and professional discourse in the biomedical and healthcare domains.

Verbs like `bring about`, `spark`, `be responsible for`, `foster` were excluded due to potential ambiguity, not health-related orientation, and high emotional connotation. Terms like `trigger` or `bring about` are frequently used in colloquial or emotional contexts, which can lead LLMs to hallucinate "event-based" rather than "variable-based" causal links.

The choice to fix the number of prompts at seven is empirically grounded in the requirements of majority vote aggregation. An odd number of prompts ($N = 7$) was selected to ensure a clear majority threshold ($k \geq 4$). This avoids the need for arbitrary tie-breaking rules, which would be necessary if we had used an even-numbered set. Our sensitivity analysis regarding the thresholding shows that the structural accuracy (SHD) becomes stable once 4 or 5 verbs agree on an edge. Therefore, using seven verbs provides an ideal balance between linguistic variety and computational efficiency.

The following subsections A and B provide details of our investigation into the two aforementioned sub-questions

A. Do Subtle Changes in Phrasing in the LLM Prompts Lead to Significantly Different Causal Models, and to What Extent Does Linguistic Variability Influence the Causal Relationships in the Models Inferred by LLMs?

We first applied Friedman's chi-square test [46] to the probability outputs for all causal linking verbs across each dataset to evaluate whether the model is sensitive to verb phrasing overall. This non-parametric test assesses whether multiple related samples (distributions for different causal linking verbs applied to the identical node pairs) come from the same population. The Friedman test is a non-parametric alternative to repeated-measures ANOVA. It is specifically suited for situations where the data consists of multiple related groups. We chose this test because:

- It does not require the assumption of normally distributed data.
- It is appropriate for comparing three or more related samples.
- It allows us to assess whether the model's outputs are statistically distinguishable based on verb phrasing, revealing potential verb sensitivity.

The Friedman test revealed statistically significant differences (p-value below 0.05) in output distributions depending on the causal linking verbs used:

- *Asia:* p-value $= 0.0021$
- *Cancer:* p-value $= 6.06 \times 10^{-5}$
- *Medical Diagnostic:* p-value $= 2.16 \times 10^{-9}$
- *Covid-19:* p-value $= 5.61 \times 10^{-6}$
- *Endorisk:* p-value $= 1.23 \times 10^{-6}$

These results demonstrate that subtle changes in the phrasing of LLM prompts can lead to significantly different DAG structures. The statistically significant differences across datasets indicate that the model's outputs are sensitive to the choice of causal linking verbs, confirming the presence of verb-specific biases. This finding suggests the need for further analysis to determine which verb pairs show the greatest differences.

B. Do Such Biases Lead to Systematic Distortion of the Derived Causal Structures?

Following the Friedman Chi-square test, we conducted pairwise comparisons between verb distributions using the Wilcoxon signed-rank test [46]. This allowed us to identify which specific causal linking verbs produce statistically significant outputs. This non-parametric test compares paired sample rankings across five benchmark datasets. Fig. 3 reports the resulting p-value for each verb pair. The p-value heatmaps summarize the statistical comparison of edge patterns generated by each verb. Low p-values (< 0.05) indicate significant differences between verbs, while p-values higher than 0.05 suggest similar structural behavior.

In simpler networks, such as the Asia, or Cancer dataset, the heatmaps are almost entirely red, suggesting similar structural behavior. This indicates that the choice of verb does not lead to

any distortion. The LLM's internal knowledge of these foundational medical relationships is robust enough to overcome different linguistic framings. However, as network complexity increases in the Covid-19 and Medical Diagnostic datasets, we observe significant distortions. In these cases, many pairwise comparisons result in low p-values (green areas), particularly for strong verbs like `cause`. For example, in the Covid-19 dataset, the graph generated by the verb `cause` was statistically different from almost all other versions. This suggests that verbs can trigger the LLM to include more aggressive or different causal links, leading to a distorted view of the ground truth if only one prompt is used. Because the LLM's training data includes both academic and non-academic sources, a prompt using `cause` may trigger a mixed causal logic. The model might pull from a corpus where the word is used loosely, leading it to include incorrect causal links that do not exist in the strict medical ground truth.

In summary, this analysis shows that the choice of causal linking verbs can significantly change the causal structure an LLM creates. The risk of distortion depends on both the specific verb used and the complexity of the medical domain. While some verbs form stable clusters and yield similar graphs, others consistently yield results that are statistically unique. Because the "best" verb may vary by case and requires deep research for every new dataset, we chose to use multiple verbs combined with an aggregation strategy. They also motivate exploring strategies, such as aggregating outputs across multiple verb formulations, to reduce bias and enhance the stability and reliability of the inferred causal structures.

VI. Aggregating Causal Edges to Mitigate Lexical Bias

Building on the observation that LLM outputs are sensitive to subtle differences in causal verb phrasing, we investigate whether aggregating outputs across multiple verb choices can mitigate lexical bias. Specifically, we test whether combining the causal edges suggested by different verbs through majority voting leads to more stable and reliable causal structures.

Two types of graphs were constructed per causal linking verbs:

- *Raw DAG:* Graphs directly built from the LLM's binary decisions, which may contain cycles and redundant relationships.
- *Pruned DAG:* These graphs were derived from the raw graphs by applying cycle removal and redundancy pruning, thus producing cleaner DAGs. This is called a post-processed graph.

To assess the robustness of causal edge predictions across different phrasings, we aggregated the LLM's binary judgments using a majority voting scheme. For each ordered variable pair (X_i, X_j), we computed the proportion of causal linking verbs for which the model predicted a directed edge. This yields a weighted adjacency matrix, where each entry reflects the consistency of the LLM's causal judgment across linguistic variations.

Across all verbs, the *Pruned DAG* consistently outperformed the *Raw DAG* in terms of SHD and F1 Score, validating the importance of post-processing to eliminate misleading relationships. This is because LLM often found mediators and tracked both direct and dependent relationships through them.

The experimental results across these five medical datasets reveal a consistent trend. Table III shows that individual causal linking verbs like `result in, raise the risk of` or `influence` can occasionally achieve high precision on specific tasks. However, their performance is highly sensitive to the domain and network complexity. In simpler networks like Cancer, the choice of linguistic framing is almost negligible; however, as the complexity increases in the Endorisk and Covid-19 datasets, the variance between verbs grows significantly. This variability suggests that no single verb can be considered a universal "gold standard" for LLM-based causal elicitation.

A critical finding across all benchmarks is the impact of structural pruning. In every instance, the Pruned DAG versions significantly outperformed their Raw counterparts, often reducing SHD in high-dimensional networks such as Endorisk. While pruning successfully eliminates redundant and cyclic edges, it occasionally introduces a slight increase in missing edges TENE if a valid causal path is misinterpreted as transitive.

One of the questions in our study is whether aggregating outputs across different causal linking verbs can mitigate lexical bias in LLM-generated causal structures. In all cases, the graph structure generated through majority voting outperformed those derived from individual prompts. This improvement can be attributed to the complementary strengths and inherent biases of different causal linking verbs, which elicit varied reasoning patterns from the LLM due to differences in semantics and linguistic framing. For instance, verbs like `cause` and `lead to` tend to prompt more direct causal assertions, while formulations such as `increase the chance of` or `raise the risk of` often yield more cautious, probabilistic judgments. Combining the outputs using majority voting helps reduce the impact of unusual results from individual prompts and makes the final structure more stable and reliable. This approach effectively mitigates errors tied to individual verbal formulations and allows consistent causal signals to emerge more clearly. Empirically, this is supported by lower SHD, TENE, and higher F1 scores in the voted graphs compared to those from single prompts (see Tables III), demonstrating the noise-reducing benefits of majority voting.

We compared the majority voting strategy with confidence-weighting based on LLM-provided confidence scores. We extracted the log-probabilities for each generated edge to represent the model's internal confidence. However, our empirical tests showed that confidence-weighted voting yielded results identical to those of simple majority voting. The lack of improvement from confidence weighting is likely due to the over-confidence bias common in LLMs. As shown in our analysis of mean probabilities (Table III), the model often assigns high log-probabilities (above 0.95) to edges, even when they are causally incorrect. Therefore, given that weighted methods did not provide a statistically significant performance boost, the simpler majority-rule approach minimizes the risk of overfitting to the specific log-probability calibrations of a single LLM version.

TABLE III
CONSOLIDATED EVALUATION OF CAUSAL LINKING VERBS ACROSS FIVE DATASETS (CANCER, ASIA,MEDICAL DIAGNOSTIC, COVID-19, AND ENDORISK). BOLD VALUES DENOTE THE BEST RESULTS WITHIN EACH DATASET. LOWER VALUES ARE BETTER FOR SHD, TENE, AND TERE; HIGHER VALUES ARE BETTER FOR F1 SCORE.

Verb	Type	SHD	SHD (norm)	TENE	TENE (norm)	TERE	TERE (norm)	F1 Score	log-Confidence
Cancer									
affect, cause, influence raise the risk of, lead to, result in	Raw	3	0.1200	0	0.0000	0	0.0000	0.7273	0.998
	Pruned	**0**	0.0000	0	0.0000	0	**0.0000**	**1.0000**	
increase the chance of	Raw	4	0.1600	0	0.0000	0	0.0000	0.6667	0.961
	Pruned	**0**	**0.0000**	0	0.0000	0	0.0000	**1.0000**	
Majority Vote Aggregation	Raw	3	0.1200	0	0.0000	0	0.0000	0.7273	
	Pruned	**0**	**0.0000**	0	0.0000	0	0.0000	**1.0000**	
Asia									
affect	Raw	14	0.2188	1	0.1250	0	0.0000	0.5000	0.956
	Pruned	11	0.1719	5	0.6250	0	0.0000	0.3529	
cause	Raw	10	0.1562	0	0.0000	0	0.0000	0.6154	0.950
	Pruned	7	0.1094	3	0.3750	0	0.0000	0.5882	
increase the chance of	Raw	13	0.2031	0	0.0000	0	0.0000	0.5517	0.940
	Pruned	9	0.1406	4	0.5000	0	0.0000	0.4706	
influence	Raw	12	0.1875	0	0.0000	0	0.0000	0.5714	0.956
	Pruned	5	0.0781	1	0.1250	0	0.0000	0.7368	
lead to	Raw	12	0.1875	0	0.0000	0	0.0000	0.5714	0.931
	Pruned	7	0.1094	3	0.3750	0	0.0000	0.5882	
raise the risk of	Raw	14	0.2188	1	0.1250	0	0.0000	0.5000	0.966
	Pruned	9	0.1406	4	0.5000	0	0.0000	0.4706	
result in	Raw	11	0.1719	0	0.0000	0	0.0000	0.5926	0.939
	Pruned	**4**	**0.0625**	**1**	**0.1250**	0	0.0000	**0.7778**	
Majority Vote Aggregation	Raw	11	0.1719	0	0.0000	0	0.0000	0.5926	
	Pruned	**4**	**0.0625**	**1**	**0.1250**	0	0.0000	**0.7778**	
Medical Diagnostic									
affect	Raw	15	0.1500	0	0.0000	0	0.0000	0.5455	0.997
	Pruned	8	0.0800	2	0.2222	0	0.0000	0.6364	
cause	Raw	16	0.1600	0	0.0000	0	0.0000	0.5294	0.005
	Pruned	8	0.0800	3	0.3333	0	0.0000	0.6000	
increase the chance of	Raw	16	0.1600	0	0.0000	0	0.0000	0.5294	0.991
	Pruned	8	0.0800	3	0.3333	0	0.0000	0.6000	
influence	Raw	17	0.1700	0	0.0000	0	0.0000	0.5143	0.999
	Pruned	10	0.1000	4	0.4444	0	0.0000	0.5000	
lead to	Raw	16	0.1600	0	0.0000	0	0.0000	0.5294	0.990
	Pruned	8	0.0800	2	0.2222	0	0.0000	0.6364	
raise the risk of	Raw	16	0.1600	0	0.0000	0	0.0000	0.5294	0.983
	Pruned	**6**	**0.0600**	**1**	**0.1111**	0	0.0000	**0.7273**	
result in	Raw	17	0.1700	0	0.0000	0	0.0000	0.5143	0.998
	Pruned	10	0.1000	4	0.4444	0	0.0000	0.5000	
Majority Vote Aggregation	Raw	15	0.1500	0	0.0000	0	0.0000	0.5455	
	Pruned	**6**	**0.0600**	**1**	**0.1111**	0	0.0000	**0.7273**	
Covid-19									
affect	Raw	18	0.1488	3	0.1875	1	0.0625	0.5909	0.978
	Pruned	15	0.1240	7	0.4375	**0**	**0.0000**	0.5455	
cause	Raw	17	0.1405	4	0.2500	1	0.0625	0.5854	0.976
	Pruned	17	0.1405	9	0.5625	1	0.0625	0.4516	
increase the chance of	Raw	18	0.1488	3	0.1875	1	0.0625	0.5909	0.977
	Pruned	16	0.1322	7	0.4375	**0**	**0.0000**	0.5294	
influence	Raw	18	0.1488	4	0.2500	1	0.0625	0.5714	0.985
	Pruned	14	0.1157	8	0.5000	1	0.0625	0.5333	
lead to	Raw	17	0.1405	3	0.1875	1	0.0625	0.6047	0.983
	Pruned	15	0.1240	8	0.5000	1	0.0625	0.5161	
raise the risk of	Raw	22	0.1818	2	0.1250	2	0.1250	0.5600	0.976
	Pruned	11	0.0909	6	0.3750	2	0.1250	0.6452	
result in	Raw	12	0.0992	2	0.1250	**0**	**0.0000**	**0.7000**	0.981
	Pruned	**10**	**0.0826**	6	0.3750	**0**	**0.0000**	0.6667	
Majority Vote Aggregation	Raw	15	0.1240	2	0.1250	6	0.3750	**0.6512**	
	Pruned	13	0.1074	5	0.3125	4	0.2500	0.6286	
Endorisk									
affect	Raw	59	0.1821	6	0.2727	1	0.0455	0.3516	0.972
	Pruned	25	0.0772	8	0.3636	1	0.0455	0.5283	
cause	Raw	47	0.1451	12	0.5455	**0**	**0.0000**	0.2985	0.984
	Pruned	32	0.0988	14	0.6364	**0**	**0.0000**	0.3333	
increase the chance of	Raw	63	0.1944	6	0.2727	3	0.1364	0.3368	0.963
	Pruned	38	0.1173	12	0.5455	3	0.1364	0.3448	
influence	Raw	57	0.1759	3	0.1364	**0**	**0.0000**	0.4000	0.972
	Pruned	17	0.0525	5	0.2273	**0**	**0.0000**	0.6667	
lead to	Raw	50	0.1543	5	0.2273	**0**	**0.0000**	0.4048	0.977
	Pruned	22	0.0679	7	0.3182	**0**	**0.0000**	0.5769	
raise the risk of	Raw	65	0.2006	4	0.1818	1	0.0455	0.3564	0.966
	Pruned	27	0.0833	8	0.3636	1	0.0455	0.5091	
result in	Raw	49	0.1512	6	0.2727	**0**	**0.0000**	0.3951	0.976
	Pruned	21	0.0648	8	0.3636	**0**	**0.0000**	0.5714	
Majority Vote Aggregation	Raw	49	0.1512	**1**	**0.0455**	2	0.0909	0.4615	
	Pruned	**16**	**0.0494**	3	0.1364	2	0.0909	**0.7037**	

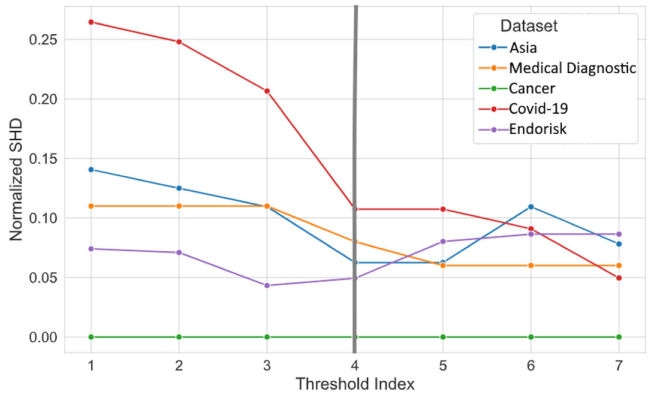


Fig. 4. Sensitivity analysis of the majority voting threshold across five medical benchmark datasets. The Normalized Structural Hamming Distance (SHD) is plotted against the threshold index (k), where an edge is included if at least k out of 7 causal linking verbs agree.

Additionally, we performed a sensitivity analysis on the majority voting threshold (the minimum number of verbs that must agree for an edge to be included). As shown in Fig. 4, the performance varies significantly depending on the threshold and the dataset's complexity. It shows an improvement as the threshold moves from one to four (majority). A threshold of one is too permissive, capturing "linguistic noise" and hallucinated transitive edges. A threshold of $k \geq 4$ (out of 7) consistently minimizes the SHD. While higher thresholds (6 or 7) slightly reduce SHD in the Covid-19 dataset, they risk increasing TENE (missing edges) in the Endorisk dataset by being overly conservative.This empirical evidence justifies our choice of a simple majority ($k \geq 4$) as it provides the most stable performance across both classic and contemporary medical benchmarks.

VII. Discussion and Limitations

Our findings suggest that LLMs can serve as an augmentation tool for eliciting expert knowledge and constructing causal graphs. However, our study had several limitations.

The primary limitations of this study are related to the availability of the benchmark datasets. The datasets used in this field [37] were constructed several decades ago based on clinical knowledge available at that time. Since then, medical science has advanced considerably, and additional causal relationships may now be recognized that are not reflected in the original graph structures. Consequently, some of the LLM-predicted edges marked as false positives in our evaluation may represent valid connections that have since been discovered. This highlights a significant challenge in evaluating LLM-based expert elicitation: the potential discrepancy between "ground truth" graphs and contemporary domain knowledge.

Another important consideration is the role of linguistic framing. Our experiments revealed that LLM predictions can be sensitive to the specific causal linking verbs used in the query (e.g., `cause` vs. `lead to` vs. `increase the chance of`). While the majority voting across varied verb formulations improved stability, this finding highlights the inherent ambiguity in natural language prompts. It motivates further research on prompt standardization or semantic normalization techniques for causal elicitation.

Aggregation generally provides the most robust results. However, the Covid-19 dataset highlights a limitation of ensemble methods, as the individual verb `result in` outperformed the majority-vote aggregate. This variation occurs because the optimal verb changes across domains. For instance, `raise the risk of` peaks for the Medical Diagnostic network. Because the true topology is unknown a priori in genuine discovery scenarios, a single-verb approach introduces severe high-variance risks. Majority voting serves as a crucial variance-reduction mechanism that stabilizes performance and cancels out single-verb errors. This strategy becomes vital in larger topologies like Endorisk, where phrasings introduce heavy linguistic noise and false transitive edges.

Finally, the proposed LLM-based causal discovery framework is characterized by certain scalability constraints. Because each pair is evaluated directionally, the total number of LLM queries per causal verb equals $n(n-1)$. Therefore, the computational complexity of the discovery process grows as ($O(n^2)$). Hence, the total API expenditure also grows quadratically with n.

For the GPT-4o-mini model, pricing (as of late 2024) is approximately \$0.15 per million input tokens and \$0.60 per million output tokens. In the current setup, each query contains approximately 150–250 input tokens and produces a single output token ("True" or "False"). Given these numbers, the estimated cost per query is approximately \$0.000025–\$0.00004. Thus, the per-query cost is negligible; however, the cumulative cost increases with the total number of pairwise evaluations. Estimated total costs under quadratic scaling of models are summarized below:

- Small to moderate scale ($n, < 18$, 306 queries): approximately \$0.01–0.02. This scale is ideal for rapid experimentation and pilot studies.
- Moderate to large scale ($n \approx 50$, ∼2,500 queries): approximately \$0.06–0.10. At this level, the approach remains highly affordable for most research budgets.
- Large scale ($n \approx 100$, ∼10,000 queries): approximately \$0.25–0.40. While the financial cost remains manageable, this scale marks the transition point where temporal latency becomes a more significant constraint than budget.

Overall, GPT-4o-mini is economically suitable for small-to-moderate causal discovery tasks. However, for very large variable sets, optimization strategies are necessary to maintain computational and financial efficiency.

To improve scalability, several strategies can be applied:

- Assuming graph sparsity, limiting the number of candidate parents per variable.
- Parallel execution of API calls, while respecting rate limits.
- Filtering nodes prior to domain knowledge to exclude improbable pairs before querying the LLM

VIII. Conclusion

This paper presents a novel approach to augmenting expert elicitation for causal modelling by integrating LLMs. The key contributions of our study include:

- A prompt-based framework for querying LLMs about pairwise causal relations and averaging their outputs,
- A systematic investigation of linguistic sensitivity in causal phrasing,
- Empirical validation using structural metrics (SHD, TENE, TERE) against known causal graphs.

To demonstrate our method, we applied it to benchmark causal graphs from the field of medical diagnostics. We demonstrated that LLMs can infer plausible causal relationships between variables, yielding graph structures that align reasonably with established ground truths across five benchmark healthcare datasets: Asia, Cancer, Medical Diagnostic, Covid-19, and Endorisk. We investigated how LLMs can support causal structure discovery across different linguistic formulations, particularly focusing on the effect of using multiple causal linking verbs. We proposed aggregating the results via majority voting across seven semantically distinct prompts to mitigate linguistic bias and reduce variability introduced by prompt phrasing.

The present work focuses strictly on validating the performance and robustness of the automated LLM-based elicitation and aggregation pipeline, without any expert-in-the-loop revision or feedback. Thus, these findings lay a strong foundation for future human-in-the-loop systems, in which LLMs would serve as a first-pass inference engine, and domain experts would subsequently evaluate, validate, or revise the suggested structures before deployment in clinical decision support. This approach may mitigate the risks associated with hallucinations in LLMs. It can also address potential biases in LLM-generated outputs [47], promoting fairness in high-stakes domains such as diagnostics, treatment planning, and policy-making.

Future research should expand on this methodology in several key areas to improve the reliability of causal discovery in healthcare. First, it is important to test this approach across a wider range of LLMs. This should include smaller, specialized models (such as BioGPT or ClinicalBERT) to see if they perform as well as larger. Additionally, it may be beneficial to apply different fin-tuning methods. We also plan to investigate the use of Retrieval-Augmented Generation (RAG) [48]. RAG allows a model to look up relevant documents or evidence from external sources during the reasoning process. Such capability is important in the clinical domain, where information must be accurate, up-to-date, and evidence-based. RAG-enabled models allow for integrating external medical knowledge sources, such as curated clinical guidelines, electronic health record repositories, and biomedical literature, and therefore, produce outputs that are more relevant, factually accurate, and context-aware [49], [50]. This approach can reduce hallucination risks and increase transparency and trustworthiness in clinical recommendations. Additionally, combining RAG with structured medical ontologies can improve interpretability and enable explainable reasoning.

VIII. Supplementary Data

The code created for this study can be accessed via the provided GitHub repository link: https://github.com/ExcellentDarkTea/LLM-Causal-Discovery

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